

Final Project Causal

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```
library(fixest)

## Warning: package 'fixest' was built under R version 4.5.3
library(did)

## Warning: package 'did' was built under R version 4.5.3
library(bacondecomp)

## Warning: package 'bacondecomp' was built under R version 4.5.3
library(dplyr)

## Warning: package 'dplyr' was built under R version 4.5.3
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
library(ggplot2)

## Warning: package 'ggplot2' was built under R version 4.5.3
library(tidyr)
library(modelsummary)

## Warning: package 'modelsummary' was built under R version 4.5.3
library(patchwork)

## Warning: package 'patchwork' was built under R version 4.5.3
```

Preparing data

```
df <- read.csv("C:/Users/jegha/OneDrive/Desktop/URI M1 S2/STA592 Causal/simulated_panel_data.csv")

df$cohort      <- ifelse(is.infinite(df$adoption_time), 0, df$adoption_time)
df$event_time  <- ifelse(is.infinite(df$adoption_time), NA,
                        df$time - df$adoption_time)
```

```

print(paste0("Units:  ", n_distinct(df$id)))

## [1] "Units:  100"
print(paste0("Periods: ", n_distinct(df$time)))

## [1] "Periods: 100"
print(table(df$cohort))

##
##    0    40    60
## 4000 3000 3000

print(paste0("True ATT (all treated):  ", round(mean(df$Treatment_effect[df$D == 1]), 4)))

## [1] "True ATT (all treated):  2.7684"
print(paste0("True ATT cohort g=40:  ", round(mean(df$Treatment_effect[df$D == 1 & df$cohort == 40]), 4)))

## [1] "True ATT cohort g=40:  3.556"
print(paste0("True ATT cohort g=60:  ", round(mean(df$Treatment_effect[df$D == 1 & df$cohort == 60]), 4)))

## [1] "True ATT cohort g=60:  1.5966"

```

Outcome Trends by Group

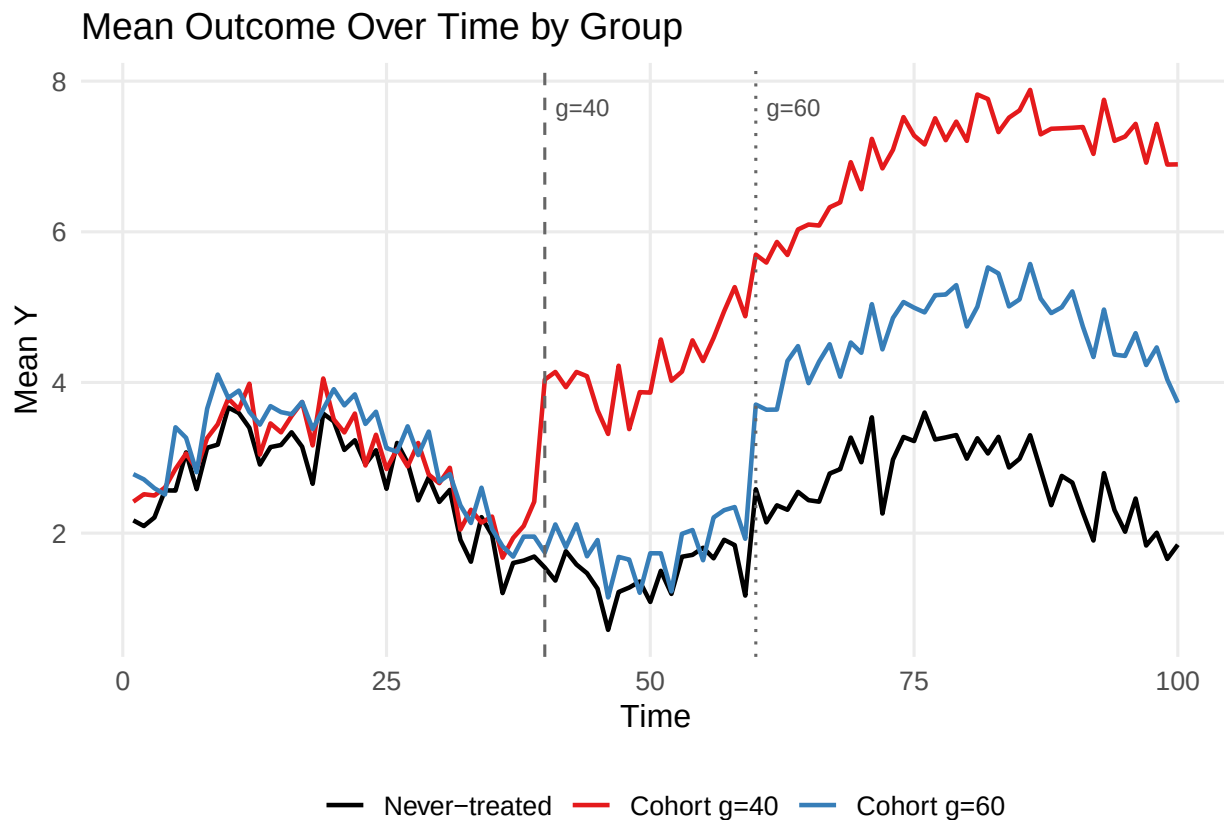
```

trend_data <- df %>%
  group_by(time, group) %>%
  summarise(Y_mean = mean(Y), .groups = "drop")

p_trends<-ggplot(trend_data, aes(x = time, y = Y_mean,
                                color = group, linetype = group)) +
  geom_line(linewidth = 0.8) +
  geom_vline(xintercept = 40, linetype = "dashed", color = "gray40") +
  geom_vline(xintercept = 60, linetype = "dotted", color = "gray40") +
  annotate("text", x = 41, y = max(trend_data$Y_mean) * 0.97,
          label = "g=40", hjust = 0, size = 3.2, color = "gray30") +
  annotate("text", x = 61, y = max(trend_data$Y_mean) * 0.97,
          label = "g=60", hjust = 0, size = 3.2, color = "gray30") +
  scale_color_manual(
    values = c("control" = "black", "treated_1" = "#E41A1C", "treated_2" = "#377EB8"),
    labels = c("Never-treated", "Cohort g=40", "Cohort g=60")) +
  scale_linetype_manual(
    values = c("control" = "solid", "treated_1" = "solid", "treated_2" = "solid"),
    labels = c("Never-treated", "Cohort g=40", "Cohort g=60")) +
  labs(title = "Mean Outcome Over Time by Group",
       x = "Time", y = "Mean Y", color = "", linetype = "") +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom", panel.grid.minor = element_blank())

p_trends

```



```
ggsave("fig_trends.png", p_trends, width = 7, height = 4)
```

TWFE Method

```
twfe <- feols(Y ~ D | id + time,
              data = df,
              vcov = ~id)

summary(twfe)

## OLS estimation, Dep. Var.: Y
## Observations: 10,000
## Fixed-effects: id: 100, time: 100
## Standard-errors: Clustered (id)
## Estimate Std. Error t value Pr(>|t|)
## D 2.2793 0.126884 17.9636 < 2.2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 1.11837 Adj. R2: 0.769039
## Within R2: 0.230156

twfe_coef <- as.numeric(coef(twfe)["D"])
twfe_se <- as.numeric(se(twfe)["D"])
true_att <- mean(df$Treatment_effect[df$D == 1])

print(paste0("TWFE estimate: ", round(twfe_coef, 4)))
```

```
## [1] "TWFE estimate: 2.2793"
print(paste0("True ATT:      ", round(true_att, 4)))

## [1] "True ATT:      2.7684"
print(paste0("Bias:         ", round(twfe_coef - true_att, 4)))

## [1] "Bias:         -0.4891"
df_es <- df %>% filter(!is.na(event_time) | group == "control")

twfe_es <- feols(Y ~ i(event_time, ref = -1) | id + time,
                data = df_es,
                vcov = ~id)

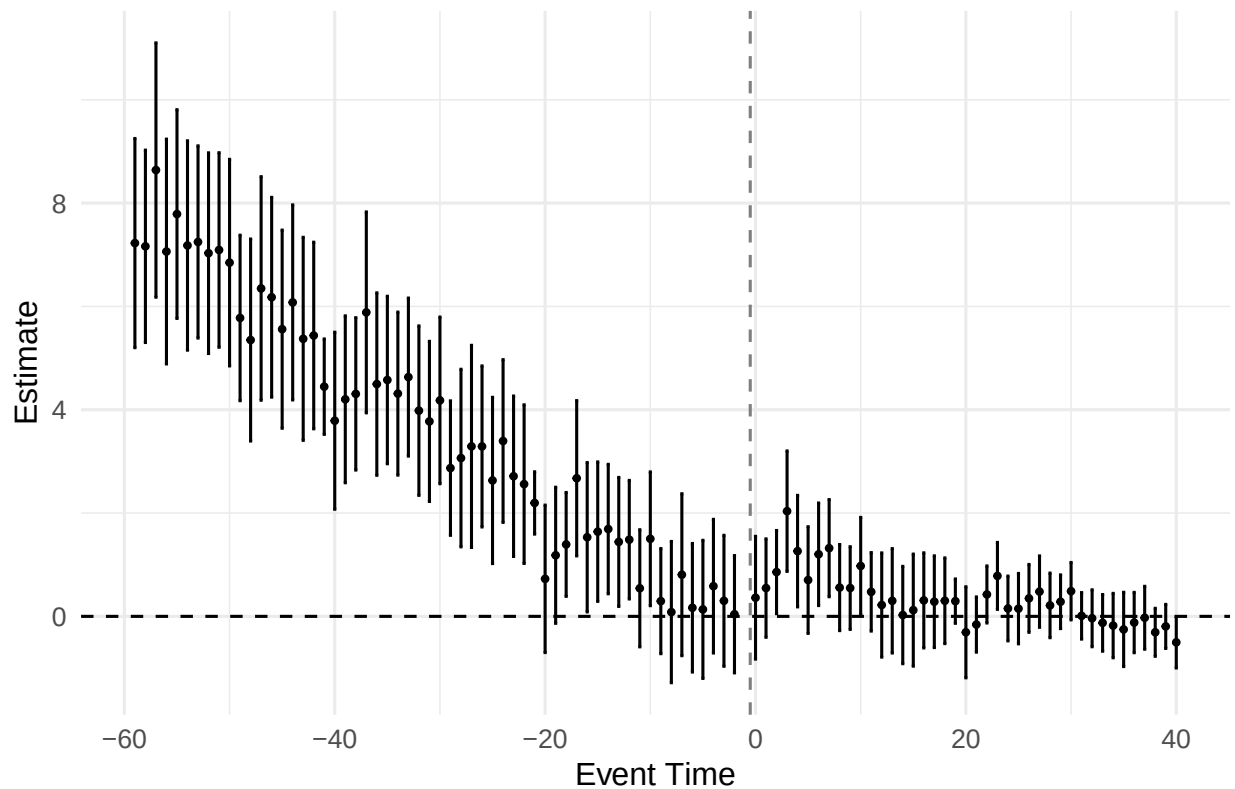
## NOTE: 4,000 observations removed because of NA values (RHS: 4,000).
## The variables 'event_time::41', 'event_time::42', 'event_time::43',
## 'event_time::44', 'event_time::45', 'event_time::46' and 14 others have been
## removed because of collinearity (see $collin.var).

twfe_es_coef <- coef(twfe_es)
twfe_es_ci <- confint(twfe_es)

twfe_es_df <- data.frame(
  event_time = as.numeric(gsub("event_time::", "", names(twfe_es_coef))),
  estimate = as.numeric(twfe_es_coef),
  conf.low = as.numeric(twfe_es_ci[, 1]),
  conf.high = as.numeric(twfe_es_ci[, 2])
)

p_twfe_es <- ggplot(twfe_es_df, aes(x = event_time, y = estimate)) +
  geom_point(size = 0.8) +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0.3) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = -0.5, linetype = "dashed", color = "gray50") +
  labs(title = "TWFE Event Study",
       x = "Event Time", y = "Estimate") +
  theme_minimal(base_size = 12)
p_twfe_es
```

TWFE Event Study



The TWFE estimate is 2.279 (SE = 0.127, $p < 0.001$), compared to the true ATT of 2.768, yielding a downward bias of -0.489 (approximately 18%). Despite being highly statistically significant, the estimate is meaningfully incorrect. This bias arises because TWFE pools all 2×2 DiD comparisons, including those that use already-treated units as controls. This problem will be addressed through Goodman-Bacon decomposition.

Goodman-Bacon Decomposition Method

```
df_bacon <- df %>%
  filter(cohort != 0) %>%
  bind_rows(df %>% filter(cohort == 0)) %>%
  select(id, time, Y, D)

bacon_out <- bacon(Y ~ D,
  data      = df_bacon,
  id_var    = "id",
  time_var  = "time")

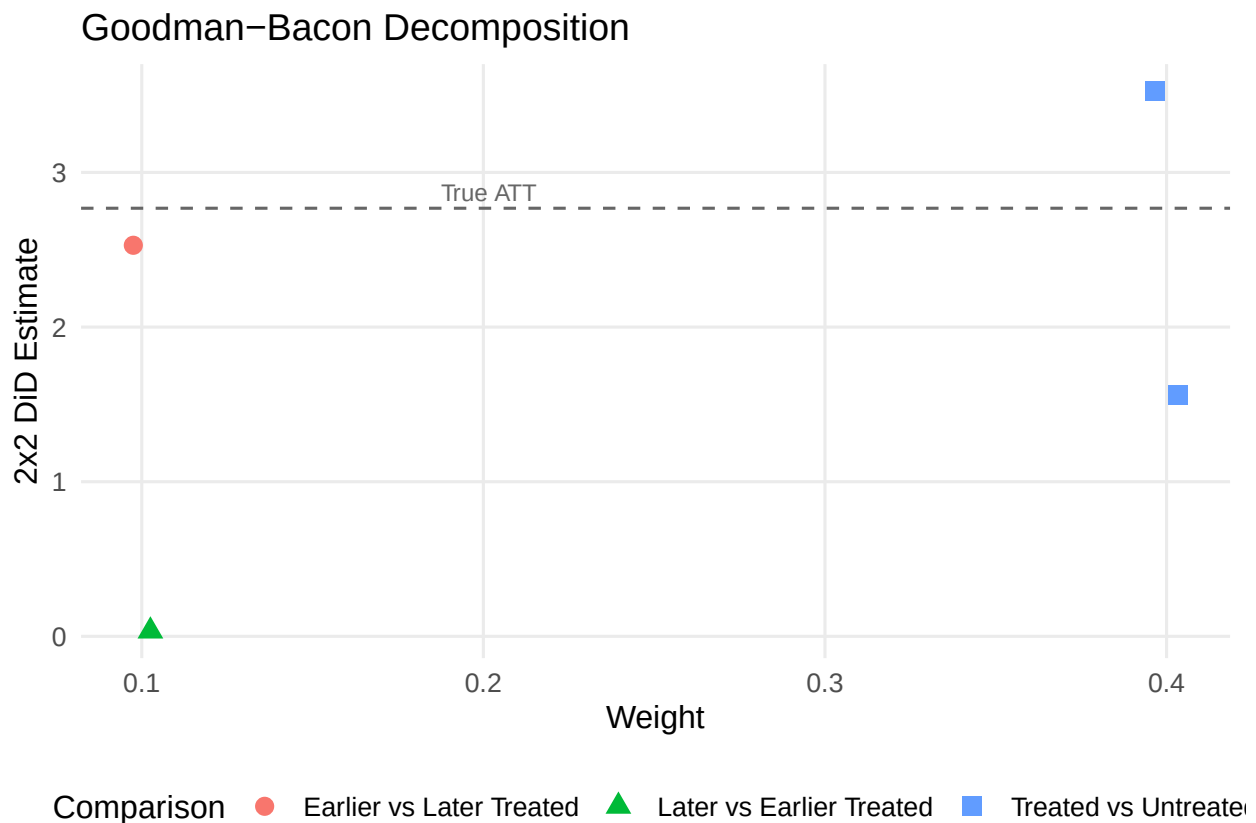
##               type weight avg_est
## 1 Earlier vs Later Treated 0.09753 2.52894
## 2 Later vs Earlier Treated 0.10253 0.03395
## 3   Treated vs Untreated 0.79993 2.53667
print(paste0("Bacon weighted avg: ", round(sum(bacon_out$estimate * bacon_out$weight), 4)))

## [1] "Bacon weighted avg: 2.2793"
```

```
print(paste0("TWFE estimate:      ", round(twfe_coef, 4)))
```

```
## [1] "TWFE estimate:      2.2793"
```

```
ggplot(bacon_out, aes(x = weight, y = estimate, shape = type, color = type)) +
  geom_point(size = 3) +
  geom_hline(yintercept = true_att, linetype = "dashed", color = "gray40") +
  annotate("text", x = max(bacon_out$weight) * 0.5,
           y = true_att + 0.1,
           label = "True ATT", size = 3.2, color = "gray40") +
  labs(title = "Goodman-Bacon Decomposition",
       x = "Weight", y = "2x2 DiD Estimate",
       shape = "Comparison", color = "Comparison") +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom", panel.grid.minor = element_blank())
```



The Goodman-Bacon decomposition breaks the TWFE estimate of 2.279 into three 2×2 comparisons. The dominant comparison is treated vs. never-treated (weight = 0.800, estimate = 2.537), which is clean and close to the true ATT. The problematic comparison is later vs. earlier treated (weight = 0.103, estimate = 0.034), where cohort $g = 60$ uses the already-treated cohort $g = 40$ as a control. This comparison severely underestimates the true effect. The earlier vs. later treated comparison (weight = 0.098, estimate = 2.529) is less distorted. The weighted average of these three components exactly recovers the TWFE estimate of 2.279, confirming that the downward bias is driven by the later vs. earlier treated comparison receiving non-trivial weight with a near-zero estimate.

Callaway-Sant'Anna Method

```
cs <- att_gt(  
  yname      = "Y",  
  tname      = "time",  
  idname     = "id",  
  gname      = "cohort",  
  xformula   = ~ x1 + x2 + x3,  
  control_group = "nevertreated",  
  data       = df,  
  est_method  = "reg"  
)
```

```
cs_overall <- aggte(cs, type = "simple")  
summary(cs_overall)
```

```
##  
## Call:  
## aggte(MP = cs, type = "simple")  
##  
## Reference: Callaway, Brantly and Pedro H.C. Sant'Anna. "Difference-in-Differences with Multiple Time  
##  
##  
##      ATT      Std. Error      [ 95%  Conf. Int.]  
##  2.3413          0.219      1.912      2.7705 *  
##  
##  
## ---  
## Signif. codes:  `*' confidence band does not cover 0  
##  
## Control Group:  Never Treated,  Anticipation Periods:  0  
## Estimation Method:  Outcome Regression
```

```
cs_cohort <- aggte(cs, type = "group")  
summary(cs_cohort)
```

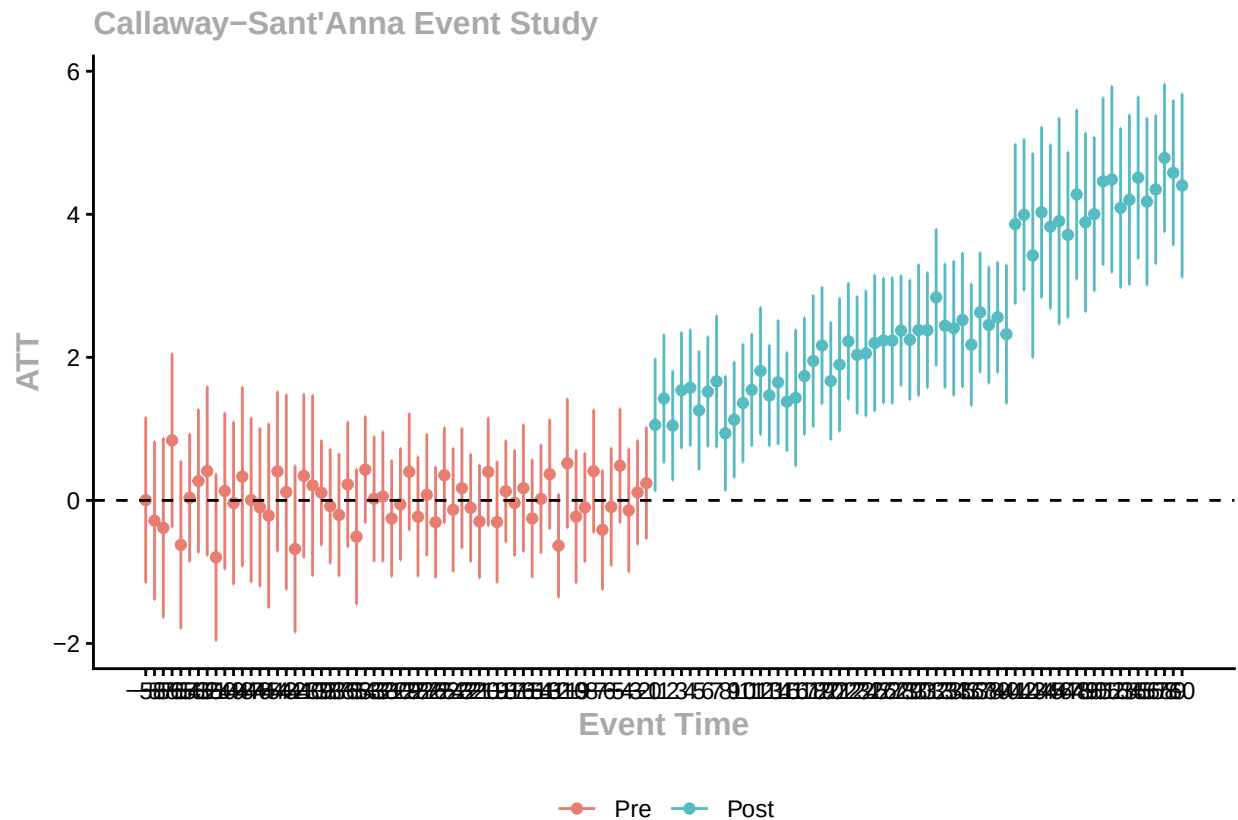
```
##  
## Call:  
## aggte(MP = cs, type = "group")  
##  
## Reference: Callaway, Brantly and Pedro H.C. Sant'Anna. "Difference-in-Differences with Multiple Time  
##  
##  
## Overall summary of ATT's based on group/cohort aggregation:  
##      ATT      Std. Error      [ 95%  Conf. Int.]  
##  2.1591          0.228      1.7121      2.606 *  
##  
##  
## Group Effects:  
## Group Estimate Std. Error [95% Simult.  Conf. Band]  
##    40    3.0883    0.2529      2.5408      3.6358 *  
##    60    1.2298    0.2502      0.6883      1.7713 *  
## ---  
## Signif. codes:  `*' confidence band does not cover 0  
##
```

```
## Control Group: Never Treated, Anticipation Periods: 0
## Estimation Method: Outcome Regression

cs_es <- aggte(cs, type = "dynamic", na.rm = TRUE)

p_cs_es <- ggdid(cs_es,
  title = "Callaway-Sant'Anna Event Study",
  xlab = "Event Time",
  ylab = "ATT")

ggsave("fig_cs_es.png", p_cs_es, width = 7, height = 4)
p_cs_es
```



The Callaway-Sant'Anna estimator, using never-treated units as the control group and conditioning on covariates x_1 , x_2 , x_3 , produces three levels of aggregation. The overall ATT is 2.341 (SE = 0.218, 95% CI: [1.915, 2.768]), close to the true value of 2.768 and a clear improvement over the TWFE estimate of 2.279. At the cohort level, the estimated ATT for cohort $g = 40$ is 3.088 (SE = 0.262) against a true value of 3.556, and for cohort $g = 60$ is 1.230 (SE = 0.234) against a true value of 1.597. Both cohort estimates are statistically significant and directionally correct, with remaining gaps attributable to finite-sample estimation error.

The event-study aggregation confirms two key features of the DGP. First, all pre-treatment coefficients (event times -58 through -1) are small and statistically indistinguishable from zero, supporting the parallel trends assumption. Second, post-treatment estimates increase monotonically with event time from 1.054 at event time 0 to 4.788 at event time 58, consistent with the linearly growing treatment effects embedded in the simulation. This dynamic pattern is entirely missed by the single TWFE coefficient, underscoring the value of the event-study approach.

Sun-Abraham Method

```
sa <- feols(Y ~ sunab(adoption_time, time) | id + time,
            data = df,
            vcov = ~id)

sa_agg <- aggregate(sa, agg = "ATT")
sa_agg

##      Estimate Std. Error  t value      Pr(>|t|)
## ATT 2.315001  0.1891938 12.23614 1.608576e-21

sa_coef <- coef(sa)
sa_ci    <- confint(sa)

sa_es_df <- data.frame(
  event_time = as.numeric(gsub("time:", "", names(sa_coef))),
  estimate    = as.numeric(sa_coef),
  conf.low    = as.numeric(sa_ci[, 1]),
  conf.high   = as.numeric(sa_ci[, 2])
)

p_sa <- ggplot(sa_es_df, aes(x = event_time, y = estimate)) +
  geom_point(size = 0.8) +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0.3) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = -0.5, linetype = "dashed", color = "gray50") +
  labs(title = "Sun-Abraham Event Study",
       x = "Event Time", y = "Estimate") +
  theme_minimal(base_size = 12)

ggsave("fig_sa_es.png", p_sa, width = 7, height = 4)
```

The Sun-Abraham estimator yields an overall ATT of 2.315 (SE = 0.189, $p < 0.001$), against a true value of 2.768, for a bias of -0.453 . The interaction-weighted event study reveals the same dynamic structure recovered by Callaway-Sant'Anna: all pre-treatment coefficients are small and most are statistically indistinguishable from zero, and post-treatment estimates grow monotonically from 1.068 at event time 0 to 4.699 at event time 58. A handful of pre-treatment coefficients reach nominal significance, but their magnitudes are small relative to the post-treatment estimates, and no systematic trend is visible, consistent with the parallel trends assumption holding in the DGP. The close agreement between the Sun-Abraham and Callaway-Sant'Anna event-study profiles cross-validates both estimators.

Overlay Event-Study Comparison

```
true_es <- df %>%
  filter(D == 1) %>%
  group_by(event_time) %>%
  summarise(estimate = mean(Treatment_effect), .groups = "drop") %>%
  mutate(method = "True ATT", conf.low = estimate, conf.high = estimate)

cs_es_df <- data.frame(
  event_time = cs_es$egt,
  estimate    = cs_es$att.egt,
```

```

    conf.low   = cs_es$att.egt - 1.96 * cs_es$se.egt,
    conf.high  = cs_es$att.egt + 1.96 * cs_es$se.egt,
    method     = "Callaway-Sant'Anna"
  )

twfe_es_coef <- coef(twfe_es)
twfe_es_ci   <- confint(twfe_es)

twfe_es_df <- data.frame(
  event_time = as.numeric(gsub("event_time::", "", names(twfe_es_coef))),
  estimate    = as.numeric(twfe_es_coef),
  conf.low    = as.numeric(twfe_es_ci[, 1]),
  conf.high   = as.numeric(twfe_es_ci[, 2]),
  method      = "TWFE"
)

sa_coef <- coef(sa)
sa_ci   <- confint(sa)

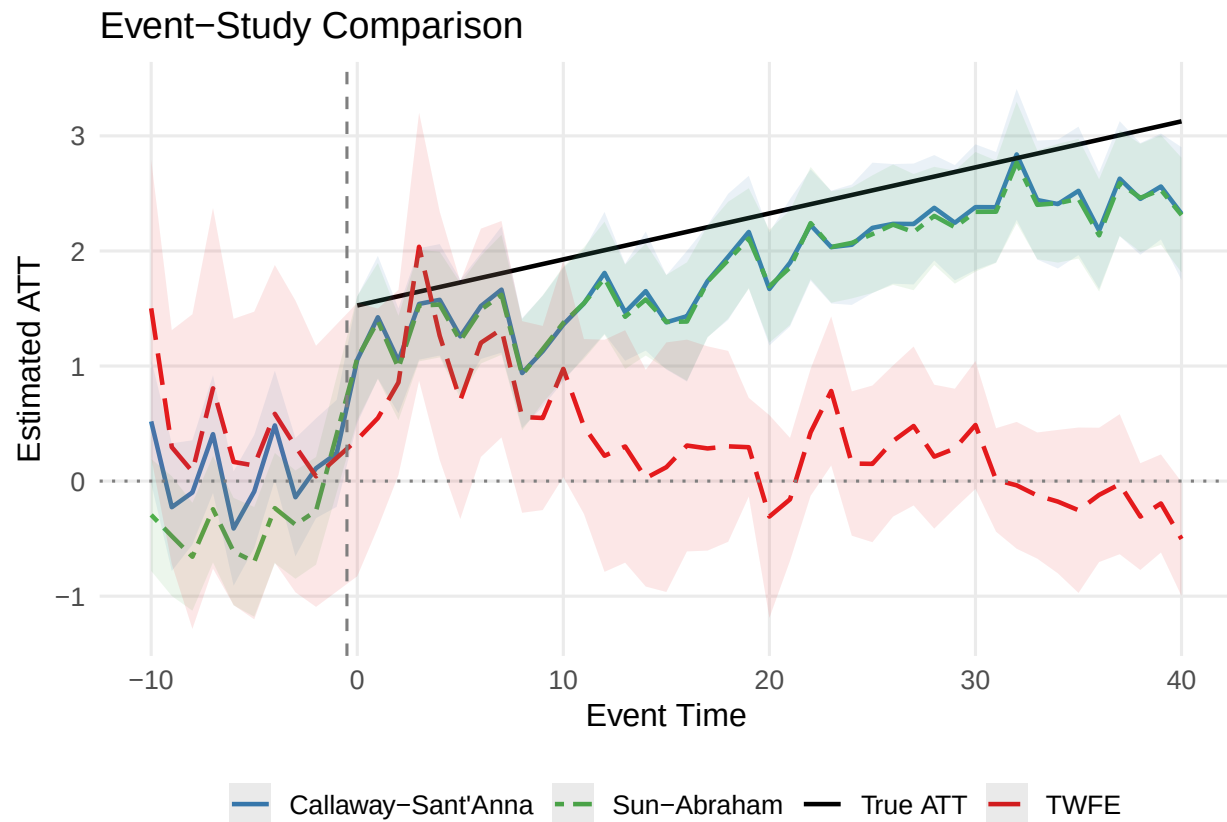
sa_es_df <- data.frame(
  event_time = as.numeric(gsub("time::", "", names(sa_coef))),
  estimate    = as.numeric(sa_coef),
  conf.low    = as.numeric(sa_ci[, 1]),
  conf.high   = as.numeric(sa_ci[, 2]),
  method      = "Sun-Abraham"
)

combined_es <- bind_rows(true_es, cs_es_df, twfe_es_df, sa_es_df) %>%
  filter(event_time >= -10, event_time <= 40)

ggplot(combined_es, aes(x = event_time, y = estimate,
                        color = method, linetype = method)) +
  geom_line(linewidth = 0.8) +
  geom_ribbon(
    aes(ymin = conf.low, ymax = conf.high, fill = method),
    alpha = 0.1, color = NA,
    data = combined_es %>% filter(method != "True ATT")
  ) +
  geom_vline(xintercept = -0.5, linetype = "dashed", color = "gray50") +
  geom_hline(yintercept = 0, linetype = "dotted", color = "gray50") +
  scale_color_manual(values = c(
    "True ATT"      = "black",
    "TWFE"          = "#E41A1C",
    "Callaway-Sant'Anna" = "#377EB8",
    "Sun-Abraham"   = "#4DAF4A")) +
  scale_linetype_manual(values = c(
    "True ATT"      = "solid",
    "TWFE"          = "longdash",
    "Callaway-Sant'Anna" = "solid",
    "Sun-Abraham"   = "twodash")) +
  scale_fill_manual(
    values = c("True ATT" = NA, "TWFE" = "#E41A1C",
              "Callaway-Sant'Anna" = "#377EB8", "Sun-Abraham" = "#4DAF4A"),
    guide = "none") +

```

```
labs(title = "Event-Study Comparison",
     x = "Event Time", y = "Estimated ATT",
     color = "", linetype = "") +
theme_minimal(base_size = 12) +
theme(legend.position = "bottom", panel.grid.minor = element_blank())
```



Summary Comparison Table

```
true_overall <- mean(df$Treatment_effect[df$D == 1])
true_g40     <- mean(df$Treatment_effect[df$D == 1 & df$cohort == 40])
true_g60     <- mean(df$Treatment_effect[df$D == 1 & df$cohort == 60])

cs_overall_est <- cs_overall$overall.att
cs_overall_se_ <- cs_overall$overall.se
cs_g40_est     <- cs_cohort$att.egt[cs_cohort$egt == 40]
cs_g60_est     <- cs_cohort$att.egt[cs_cohort$egt == 60]
cs_g40_se_     <- cs_cohort$se.egt[cs_cohort$egt == 40]
cs_g60_se_     <- cs_cohort$se.egt[cs_cohort$egt == 60]

sa_overall_est <- sa_agg["ATT", "Estimate"]
sa_overall_se_ <- sa_agg["ATT", "Std. Error"]

results_table <- data.frame(
  Method      = c("True ATT", "TWFE", "Callaway-Sant'Anna", "Sun-Abraham"),
  Overall_Est = round(c(true_overall, twfe_coef, cs_overall_est, sa_overall_est), 3),
```

```

Overall_SE = round(c(NA, twfe_se, cs_overall_se_, sa_overall_se_), 3),
Overall_Bias = round(c(0,
                        twfe_coef - true_overall,
                        cs_overall_est - true_overall,
                        sa_overall_est - true_overall), 3),
g40_Est = round(c(true_g40, NA, cs_g40_est, NA), 3),
g40_SE = round(c(NA, NA, cs_g40_se_, NA), 3),
g40_Bias = round(c(0, NA, cs_g40_est - true_g40, NA), 3),
g60_Est = round(c(true_g60, NA, cs_g60_est, NA), 3),
g60_SE = round(c(NA, NA, cs_g60_se_, NA), 3),
g60_Bias = round(c(0, NA, cs_g60_est - true_g60, NA), 3)
)

```

results_table

##	Method	Overall_Est	Overall_SE	Overall_Bias	g40_Est	g40_SE
## 1	True ATT	2.768	NA	0.000	3.556	NA
## 2	TWFE	2.279	0.127	-0.489	NA	NA
## 3	Callaway-Sant'Anna	2.341	0.219	-0.427	3.088	0.253
## 4	Sun-Abraham	2.315	0.189	-0.453	NA	NA
##	g40_Bias	g60_Est	g60_SE	g60_Bias		
## 1	0.000	1.597	NA	0.000		
## 2	NA	NA	NA	NA		
## 3	-0.468	1.230	0.25	-0.367		
## 4	NA	NA	NA	NA		

This table summarizes the ATT estimates across all three methods against the known true values. At the aggregate level, all three estimators underestimate the true ATT of 2.768, with biases of -0.489 (TWFE), -0.453 (Sun-Abraham), and -0.427 (Callaway-Sant'Anna). The differences in overall bias are modest, reflecting the fact that aggregate estimates obscure the heterogeneity across cohorts. The key distinction emerges at the cohort level, where Callaway-Sant'Anna recovers estimates of 3.088 for cohort $g = 40$ and 1.230 for cohort $g = 60$, both statistically significant and directionally correct relative to the true values of 3.556 and 1.597 respectively. TWFE and Sun-Abraham do not produce cohort-specific aggregates in this form. The remaining bias in the CS cohort estimates (-0.468 and -0.367) is attributable to finite-sample estimation error rather than a failure of identification, as the pre-trend tests confirm the parallel trends assumption holds. Overall, the table illustrates that while all methods recover the broad direction of the treatment effect, only Callaway-Sant'Anna reveals the substantial heterogeneity across adoption cohorts that TWFE obscures.

CS Cohort-Specific ATT Plot

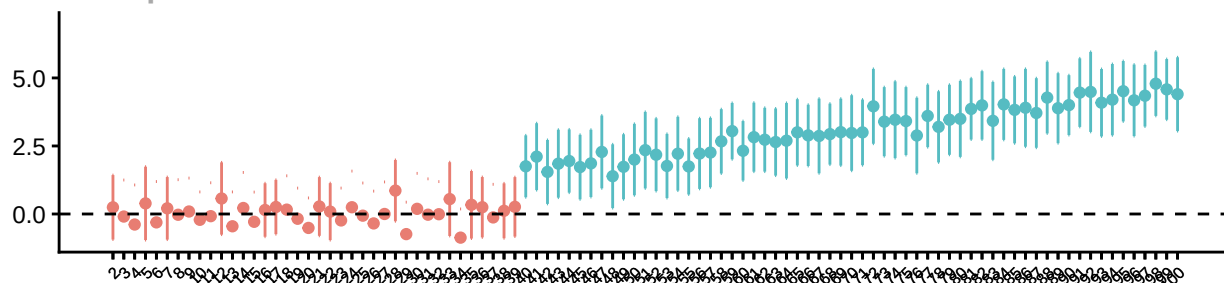
```

p_cs_cohort <- ggdid(cs, ylim = c(-1, 7),
                    title = "ATT(g,t): Callaway-Sant'Anna by Cohort") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 6))
p_cs_cohort

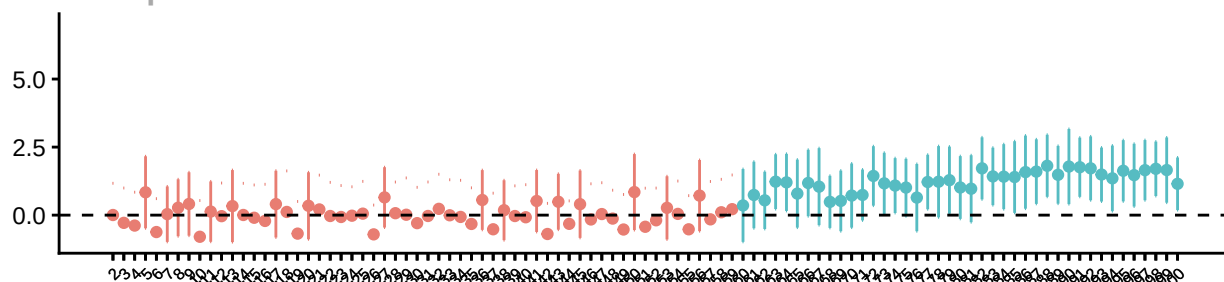
```

ATT(g,t): Callaway–Sant'Anna by Cohort

Group 40



Group 60



Pre Post

```
ggsave("fig_cs_cohort.png", p_cs_cohort, width = 9, height = 5)
```

Appendix

Callaway-Sant'Anna Method Tables

```
summary(cs)
```

```
##
## Call:
## att_gt(ymame = "Y", tname = "time", idname = "id", gname = "cohort",
##       xformula = ~x1 + x2 + x3, data = df, control_group = "nevertreated",
##       est_method = "reg")
##
## Reference: Callaway, Brantly and Pedro H.C. Sant'Anna. "Difference-in-Differences with Multiple Time
##
## Group-Time Average Treatment Effects:
##   Group Time ATT(g,t) Std. Error [95% Simult. Conf. Band]
##   40      2    0.2460    0.3401    -0.9384    1.4304
##   40      3   -0.0930    0.3853   -1.4349    1.2490
##   40      4   -0.3862    0.4175   -1.8403    1.0679
##   40      5    0.3902    0.3872   -0.9581    1.7384
##   40      6   -0.3082    0.4312   -1.8098    1.1934
##   40      7    0.2082    0.3300   -0.9409    1.3573
##   40      8   -0.0277    0.3699   -1.3158    1.2604
##   40      9    0.0920    0.3528   -1.1366    1.3206
```

##	40	10	-0.2129	0.2942	-1.2375	0.8118
##	40	11	-0.0794	0.3523	-1.3063	1.1475
##	40	12	0.5698	0.3812	-0.7578	1.8975
##	40	13	-0.4504	0.3625	-1.7128	0.8120
##	40	14	0.2206	0.3754	-1.0868	1.5280
##	40	15	-0.2861	0.3137	-1.3784	0.8063
##	40	16	0.1474	0.2822	-0.8353	1.1302
##	40	17	0.2576	0.2864	-0.7400	1.2551
##	40	18	0.1604	0.3571	-1.0831	1.4039
##	40	19	-0.1747	0.3234	-1.3011	0.9516
##	40	20	-0.5119	0.3165	-1.6139	0.5902
##	40	21	0.2774	0.3096	-0.8007	1.3554
##	40	22	0.0878	0.3006	-0.9589	1.1346
##	40	23	-0.2360	0.3413	-1.4247	0.9527
##	40	24	0.2485	0.3814	-1.0797	1.5767
##	40	25	-0.0607	0.3444	-1.2601	1.1388
##	40	26	-0.3458	0.3407	-1.5324	0.8407
##	40	27	0.0013	0.3379	-1.1754	1.1781
##	40	28	0.8601	0.3236	-0.2668	1.9869
##	40	29	-0.7389	0.3371	-1.9130	0.4351
##	40	30	0.1903	0.3744	-1.1134	1.4940
##	40	31	-0.0254	0.3783	-1.3430	1.2922
##	40	32	-0.0077	0.3443	-1.2066	1.1912
##	40	33	0.5484	0.3892	-0.8069	1.9037
##	40	34	-0.8687	0.3028	-1.9232	0.1858
##	40	35	0.3393	0.3564	-0.9020	1.5805
##	40	36	0.2501	0.3185	-0.8589	1.3591
##	40	37	-0.1254	0.3492	-1.3414	1.0907
##	40	38	0.1179	0.2898	-0.8914	1.1272
##	40	39	0.2631	0.3136	-0.8292	1.3554
##	40	40	1.7531	0.3268	0.6152	2.8910 *
##	40	41	2.1073	0.3531	0.8776	3.3371 *
##	40	42	1.5468	0.3373	0.3722	2.7214 *
##	40	43	1.8500	0.3587	0.6008	3.0992 *
##	40	44	1.9472	0.3338	0.7846	3.1097 *
##	40	45	1.7242	0.3406	0.5381	2.9103 *
##	40	46	1.8600	0.3551	0.6233	3.0968 *
##	40	47	2.2815	0.3841	0.9439	3.6191 *
##	40	48	1.3894	0.3366	0.2174	2.5615 *
##	40	49	1.7347	0.3430	0.5400	2.9293 *
##	40	50	2.0007	0.3772	0.6871	3.3144 *
##	40	51	2.3467	0.4040	0.9399	3.7535 *
##	40	52	2.1784	0.3840	0.8411	3.5157 *
##	40	53	1.7659	0.3387	0.5864	2.9454 *
##	40	54	2.2158	0.3895	0.8595	3.5721 *
##	40	55	1.7469	0.2954	0.7180	2.7758 *
##	40	56	2.2216	0.3730	0.9228	3.5204 *
##	40	57	2.2587	0.3663	0.9829	3.5344 *
##	40	58	2.6702	0.3394	1.4882	3.8521 *
##	40	59	3.0457	0.2975	2.0097	4.0816 *
##	40	60	2.3235	0.3136	1.2313	3.4157 *
##	40	61	2.8242	0.3642	1.5559	4.0925 *
##	40	62	2.7304	0.3369	1.5572	3.9037 *
##	40	63	2.6446	0.3543	1.4106	3.8785 *

##	40	64	2.6947	0.3990	1.3053	4.0841 *
##	40	65	3.0010	0.3498	1.7827	4.2193 *
##	40	66	2.8901	0.3233	1.7644	4.0159 *
##	40	67	2.8684	0.3944	1.4949	4.2418 *
##	40	68	2.9359	0.3220	1.8145	4.0572 *
##	40	69	3.0082	0.3533	1.7778	4.2386 *
##	40	70	2.9765	0.3984	1.5890	4.3639 *
##	40	71	2.9995	0.3465	1.7926	4.2063 *
##	40	72	3.9580	0.3950	2.5823	5.3336 *
##	40	73	3.3930	0.3646	2.1234	4.6626 *
##	40	74	3.4637	0.4050	2.0534	4.8741 *
##	40	75	3.4146	0.3584	2.1665	4.6627 *
##	40	76	2.8846	0.3987	1.4959	4.2732 *
##	40	77	3.6045	0.3296	2.4565	4.7525 *
##	40	78	3.2019	0.3751	1.8957	4.5081 *
##	40	79	3.4635	0.3699	2.1755	4.7515 *
##	40	80	3.4960	0.4010	2.0995	4.8925 *
##	40	81	3.8624	0.3234	2.7362	4.9885 *
##	40	82	3.9923	0.3606	2.7366	5.2480 *
##	40	83	3.4241	0.4107	1.9939	4.8543 *
##	40	84	4.0283	0.3745	2.7240	5.3326 *
##	40	85	3.8266	0.3532	2.5967	5.0565 *
##	40	86	3.9038	0.4108	2.4732	5.3344 *
##	40	87	3.7131	0.3670	2.4350	4.9913 *
##	40	88	4.2775	0.3784	2.9599	5.5952 *
##	40	89	3.8880	0.3693	2.6019	5.1742 *
##	40	90	4.0018	0.3160	2.9012	5.1023 *
##	40	91	4.4606	0.3617	3.2008	5.7203 *
##	40	92	4.4874	0.4224	3.0163	5.9584 *
##	40	93	4.0897	0.3561	2.8498	5.3296 *
##	40	94	4.2025	0.3761	2.8926	5.5124 *
##	40	95	4.5116	0.3188	3.4014	5.6218 *
##	40	96	4.1777	0.3776	2.8627	5.4926 *
##	40	97	4.3465	0.3250	3.2146	5.4784 *
##	40	98	4.7882	0.3395	3.6059	5.9706 *
##	40	99	4.5801	0.3202	3.4650	5.6952 *
##	40	100	4.4025	0.3895	3.0460	5.7590 *
##	60	2	0.0038	0.3348	-1.1622	1.1697
##	60	3	-0.2837	0.3676	-1.5638	0.9965
##	60	4	-0.3841	0.3522	-1.6105	0.8424
##	60	5	0.8391	0.3799	-0.4840	2.1621
##	60	6	-0.6218	0.3524	-1.8489	0.6052
##	60	7	0.0363	0.2926	-0.9827	1.0554
##	60	8	0.2703	0.3005	-0.7763	1.3170
##	60	9	0.4098	0.3347	-0.7557	1.5753
##	60	10	-0.7957	0.3827	-2.1283	0.5369
##	60	11	0.1304	0.3185	-0.9787	1.2396
##	60	12	-0.0384	0.3498	-1.2567	1.1800
##	60	13	0.3323	0.3802	-0.9917	1.6563
##	60	14	0.0054	0.3342	-1.1586	1.1693
##	60	15	-0.0968	0.3460	-1.3019	1.1083
##	60	16	-0.2131	0.3866	-1.5595	1.1333
##	60	17	0.4049	0.3522	-0.8215	1.6313
##	60	18	0.1166	0.4330	-1.3912	1.6243

##	60	19	-0.6793	0.3393	-1.8608	0.5021
##	60	20	0.3434	0.3535	-0.8878	1.5745
##	60	21	0.2090	0.3619	-1.0514	1.4695
##	60	22	-0.0322	0.3549	-1.2682	1.2038
##	60	23	-0.0679	0.3316	-1.2225	1.0868
##	60	24	-0.0217	0.3053	-1.0850	1.0416
##	60	25	0.0544	0.3405	-1.1314	1.2402
##	60	26	-0.7073	0.3090	-1.7835	0.3690
##	60	27	0.6515	0.3184	-0.4572	1.7602
##	60	28	0.0713	0.3286	-1.0729	1.2155
##	60	29	0.0139	0.3885	-1.3390	1.3668
##	60	30	-0.2942	0.3779	-1.6103	1.0218
##	60	31	-0.0313	0.3586	-1.2802	1.2177
##	60	32	0.2302	0.3652	-1.0415	1.5020
##	60	33	-0.0057	0.3748	-1.3109	1.2995
##	60	34	-0.0665	0.3864	-1.4123	1.2792
##	60	35	-0.3244	0.3813	-1.6524	1.0036
##	60	36	0.5564	0.3152	-0.5411	1.6539
##	60	37	-0.5203	0.3810	-1.8470	0.8064
##	60	38	0.1804	0.3161	-0.9206	1.2813
##	60	39	-0.0330	0.3183	-1.1415	1.0755
##	60	40	-0.0780	0.3435	-1.2742	1.1183
##	60	41	0.5192	0.3266	-0.6182	1.6566
##	60	42	-0.6934	0.3234	-1.8195	0.4328
##	60	43	0.4915	0.2950	-0.5357	1.5188
##	60	44	-0.3200	0.2421	-1.1632	0.5231
##	60	45	0.4019	0.3554	-0.8358	1.6396
##	60	46	-0.1596	0.3765	-1.4706	1.1515
##	60	47	0.0390	0.3305	-1.1120	1.1900
##	60	48	-0.1272	0.2996	-1.1705	0.9162
##	60	49	-0.5288	0.3695	-1.8155	0.7580
##	60	50	0.8473	0.4009	-0.5488	2.2433
##	60	51	-0.4281	0.4073	-1.8465	0.9903
##	60	52	-0.1894	0.3399	-1.3730	0.9942
##	60	53	0.2675	0.3334	-0.8935	1.4284
##	60	54	0.0449	0.3466	-1.1621	1.2519
##	60	55	-0.5206	0.3549	-1.7566	0.7154
##	60	56	0.7179	0.3757	-0.5904	2.0262
##	60	57	-0.1553	0.3997	-1.5472	1.2367
##	60	58	0.1049	0.3476	-1.1056	1.3154
##	60	59	0.2187	0.3609	-1.0383	1.4756
##	60	60	0.3553	0.3840	-0.9821	1.6926
##	60	61	0.7402	0.3512	-0.4828	1.9633
##	60	62	0.5429	0.3021	-0.5092	1.5951
##	60	63	1.2290	0.2892	0.2219	2.2361 *
##	60	64	1.2032	0.3017	0.1526	2.2538 *
##	60	65	0.7907	0.3591	-0.4597	2.0412
##	60	66	1.1807	0.3505	-0.0400	2.4015
##	60	67	1.0461	0.4064	-0.3690	2.4613
##	60	68	0.4877	0.2752	-0.4706	1.4460
##	60	69	0.5224	0.3201	-0.5923	1.6372
##	60	70	0.7189	0.3385	-0.4598	1.8976
##	60	71	0.7408	0.2687	-0.1951	1.6766
##	60	72	1.4401	0.3137	0.3477	2.5325 *

```
##      60      73      1.1678      0.3230      0.0428      2.2927 *
##      60      74      1.0859      0.2893      0.0783      2.0935 *
##      60      75      1.0128      0.3033     -0.0436      2.0692
##      60      76      0.6443      0.3550     -0.5919      1.8805
##      60      77      1.2174      0.2877      0.2156      2.2192 *
##      60      78      1.2284      0.3757     -0.0801      2.5369
##      60      79      1.2846      0.3545      0.0501      2.5192 *
##      60      80      1.0154      0.3310     -0.1372      2.1681
##      60      81      0.9704      0.3532     -0.2596      2.2004
##      60      82      1.7187      0.3274      0.5785      2.8590 *
##      60      83      1.4222      0.3036      0.3650      2.4794 *
##      60      84      1.4131      0.3429      0.2191      2.6072 *
##      60      85      1.3998      0.3781      0.0830      2.7166 *
##      60      86      1.5805      0.3859      0.2367      2.9244 *
##      60      87      1.6000      0.3407      0.4136      2.7863 *
##      60      88      1.8136      0.3274      0.6734      2.9539 *
##      60      89      1.4815      0.3049      0.4198      2.5432 *
##      60      90      1.7847      0.3967      0.4031      3.1663 *
##      60      91      1.7601      0.3126      0.6716      2.8487 *
##      60      92      1.7194      0.3370      0.5458      2.8931 *
##      60      93      1.4914      0.2851      0.4986      2.4842 *
##      60      94      1.3513      0.3464      0.1449      2.5578 *
##      60      95      1.6299      0.3231      0.5045      2.7552 *
##      60      96      1.4692      0.3314      0.3151      2.6232 *
##      60      97      1.6519      0.3178      0.5452      2.7587 *
##      60      98      1.7024      0.2873      0.7019      2.7029 *
##      60      99      1.6565      0.3443      0.4575      2.8555 *
##      60     100      1.1511      0.2778      0.1836      2.1187 *
```

```
## ---
```

```
## Signif. codes: '*' confidence band does not cover 0
```

```
##
```

```
## P-value for pre-test of parallel trends assumption: 0
```

```
## Control Group: Never Treated, Anticipation Periods: 0
```

```
## Estimation Method: Outcome Regression
```

```
summary(cs_es)
```

```
##
```

```
## Call:
```

```
## aggte(MP = cs, type = "dynamic", na.rm = TRUE)
```

```
##
```

```
## Reference: Callaway, Brantly and Pedro H.C. Sant'Anna. "Difference-in-Differences with Multiple Time
```

```
##
```

```
##
```

```
## Overall summary of ATT's based on event-study/dynamic aggregation:
```

```
##      ATT      Std. Error      [ 95% Conf. Int.]
## 2.6375      0.1863      2.2724      3.0026 *
```

```
##
```

```
##
```

```
## Dynamic Effects:
```

```
##      Event time Estimate Std. Error [95% Simult. Conf. Band]
##          -58      0.0038      0.3515      -1.1532      1.1607
##          -57     -0.2837      0.3363      -1.3905      0.8231
##          -56     -0.3841      0.3816      -1.6400      0.8718
##          -55      0.8391      0.3690      -0.3756      2.0538
```

##	-54	-0.6218	0.3559	-1.7932	0.5495
##	-53	0.0363	0.2713	-0.8567	0.9294
##	-52	0.2703	0.3043	-0.7314	1.2720
##	-51	0.4098	0.3592	-0.7726	1.5922
##	-50	-0.7957	0.3548	-1.9636	0.3722
##	-49	0.1304	0.3320	-0.9622	1.2231
##	-48	-0.0384	0.3457	-1.1761	1.0994
##	-47	0.3323	0.3811	-0.9222	1.5867
##	-46	0.0054	0.3488	-1.1427	1.1534
##	-45	-0.0968	0.3364	-1.2042	1.0106
##	-44	-0.2131	0.3913	-1.5011	1.0749
##	-43	0.4049	0.3396	-0.7128	1.5226
##	-42	0.1166	0.4140	-1.2460	1.4792
##	-41	-0.6793	0.3544	-1.8457	0.4871
##	-40	0.3434	0.3478	-0.8013	1.4880
##	-39	0.2090	0.3834	-1.0529	1.4709
##	-38	0.1069	0.2231	-0.6273	0.8412
##	-37	-0.0804	0.2431	-0.8806	0.7198
##	-36	-0.2039	0.2606	-1.0618	0.6539
##	-35	0.2223	0.2664	-0.6547	1.0993
##	-34	-0.5077	0.2871	-1.4527	0.4373
##	-33	0.4299	0.2262	-0.3148	1.1746
##	-32	0.0218	0.2652	-0.8512	0.8948
##	-31	0.0529	0.2762	-0.8563	0.9622
##	-30	-0.2535	0.2466	-1.0654	0.5583
##	-29	-0.0553	0.2376	-0.8372	0.7266
##	-28	0.4000	0.2479	-0.4160	1.2161
##	-27	-0.2280	0.2545	-1.0659	0.6098
##	-26	0.0770	0.2585	-0.7738	0.9279
##	-25	-0.3053	0.2354	-1.0800	0.4695
##	-24	0.3519	0.2032	-0.3168	1.0206
##	-23	-0.1314	0.2620	-0.9936	0.7309
##	-22	0.1704	0.2552	-0.6697	1.0105
##	-21	-0.1039	0.2292	-0.8582	0.6504
##	-20	-0.2949	0.2404	-1.0861	0.4963
##	-19	0.3983	0.2302	-0.3593	1.1558
##	-18	-0.3028	0.2575	-1.1503	0.5448
##	-17	0.1278	0.2161	-0.5836	0.8392
##	-16	-0.0358	0.2246	-0.7751	0.7035
##	-15	0.1706	0.2698	-0.7175	1.0587
##	-14	-0.2527	0.2507	-1.0778	0.5724
##	-13	0.0202	0.2305	-0.7384	0.7787
##	-12	0.3664	0.2326	-0.3991	1.1320
##	-11	-0.6339	0.2196	-1.3568	0.0890
##	-10	0.5188	0.2745	-0.3846	1.4221
##	-9	-0.2267	0.2832	-1.1590	0.7055
##	-8	-0.0985	0.2312	-0.8595	0.6624
##	-7	0.4079	0.2612	-0.4518	1.2676
##	-6	-0.4119	0.2537	-1.2468	0.4231
##	-5	-0.0907	0.2509	-0.9164	0.7350
##	-4	0.4840	0.2426	-0.3145	1.2824
##	-3	-0.1403	0.2622	-1.0035	0.7228
##	-2	0.1114	0.2214	-0.6172	0.8400
##	-1	0.2409	0.2371	-0.5395	1.0212

##	0	1.0542	0.2812	0.1286	1.9797 *
##	1	1.4238	0.2720	0.5284	2.3191 *
##	2	1.0449	0.2329	0.2782	1.8115 *
##	3	1.5395	0.2472	0.7259	2.3531 *
##	4	1.5752	0.2471	0.7619	2.3884 *
##	5	1.2575	0.2518	0.4286	2.0863 *
##	6	1.5204	0.2342	0.7494	2.2913 *
##	7	1.6638	0.2796	0.7435	2.5842 *
##	8	0.9386	0.2438	0.1362	1.7409 *
##	9	1.1286	0.2461	0.3184	1.9387 *
##	10	1.3598	0.2518	0.5310	2.1886 *
##	11	1.5438	0.2382	0.7596	2.3279 *
##	12	1.8092	0.2703	0.9195	2.6990 *
##	13	1.4669	0.2151	0.7588	2.1749 *
##	14	1.6508	0.2639	0.7823	2.5193 *
##	15	1.3799	0.2093	0.6910	2.0688 *
##	16	1.4329	0.2899	0.4786	2.3873 *
##	17	1.7380	0.2486	0.9199	2.5562 *
##	18	1.9493	0.2797	1.0288	2.8698 *
##	19	2.1651	0.2489	1.3458	2.9845 *
##	20	1.6695	0.2504	0.8454	2.4936 *
##	21	1.8973	0.2826	0.9671	2.8275 *
##	22	2.2246	0.2467	1.4126	3.0366 *
##	23	2.0334	0.2500	1.2104	2.8563 *
##	24	2.0539	0.2670	1.1752	2.9326 *
##	25	2.2004	0.2891	1.2487	3.1521 *
##	26	2.2353	0.2652	1.3625	3.1081 *
##	27	2.2342	0.2681	1.3518	3.1165 *
##	28	2.3747	0.2339	1.6048	3.1447 *
##	29	2.2448	0.2555	1.4037	3.0860 *
##	30	2.3806	0.2787	1.4633	3.2979 *
##	31	2.3798	0.2449	1.5737	3.1859 *
##	32	2.8387	0.2904	1.8829	3.7945 *
##	33	2.4422	0.2637	1.5742	3.3102 *
##	34	2.4075	0.2858	1.4670	3.3481 *
##	35	2.5222	0.2848	1.5850	3.4595 *
##	36	2.1769	0.2588	1.3250	3.0288 *
##	37	2.6282	0.2549	1.7892	3.4672 *
##	38	2.4522	0.2476	1.6371	3.2672 *
##	39	2.5600	0.2349	1.7869	3.3331 *
##	40	2.3236	0.2950	1.3525	3.2946 *
##	41	3.8624	0.3393	2.7455	4.9793 *
##	42	3.9923	0.3222	2.9319	5.0527 *
##	43	3.4241	0.4344	1.9943	4.8539 *
##	44	4.0283	0.3630	2.8334	5.2233 *
##	45	3.8266	0.3483	2.6802	4.9730 *
##	46	3.9038	0.4381	2.4619	5.3457 *
##	47	3.7131	0.3522	2.5539	4.8723 *
##	48	4.2775	0.3602	3.0921	5.4630 *
##	49	3.8880	0.3803	2.6362	5.1398 *
##	50	4.0018	0.3272	2.9247	5.0789 *
##	51	4.4606	0.3554	3.2908	5.6304 *
##	52	4.4874	0.3959	3.1844	5.7903 *
##	53	4.0897	0.3386	2.9751	5.2043 *

```
##          54  4.2025    0.3614    3.0131    5.3919 *
##          55  4.5116    0.3439    3.3797    5.6435 *
##          56  4.1777    0.3554    3.0079    5.3475 *
##          57  4.3465    0.3163    3.3055    5.3875 *
##          58  4.7882    0.3145    3.7529    5.8235 *
##          59  4.5801    0.3083    3.5654    5.5949 *
##          60  4.4025    0.3900    3.1188    5.6862 *
## ---
## Signif. codes:  `*' confidence band does not cover 0
##
## Control Group:  Never Treated,  Anticipation Periods:  0
## Estimation Method:  Outcome Regression
```

Sun-Abraham Method Table

```
summary(sa)
```

```
## OLS estimation, Dep. Var.: Y
## Observations: 10,000
## Fixed-effects: id: 100,  time: 100
## Standard-errors: Clustered (id)
##      Estimate Std. Error  t value  Pr(>|t|)
## time::-59 -0.140716    0.346420 -0.406200 6.8547e-01
## time::-58 -0.134963    0.344783 -0.391443 6.9631e-01
## time::-57 -0.366146    0.311478 -1.175514 2.4261e-01
## time::-56 -0.809436    0.310385 -2.607848 1.0521e-02 *
## time::-55  0.085862    0.368361  0.233091 8.1617e-01
## time::-54 -0.561208    0.291879 -1.922741 5.7386e-02 .
## time::-53 -0.529737    0.294677 -1.797689 7.5275e-02 .
## time::-52 -0.236416    0.293065 -0.806700 4.2177e-01
## time::-51  0.175879    0.312388  0.563014 5.7470e-01
## time::-50 -0.626891    0.295106 -2.124291 3.6137e-02 *
## time::-49 -0.457767    0.335195 -1.365673 1.7514e-01
## time::-48 -0.543081    0.310480 -1.749167 8.3361e-02 .
## time::-47 -0.224392    0.317553 -0.706629 4.8146e-01
## time::-46 -0.210713    0.284288 -0.741196 4.6033e-01
## time::-45 -0.317767    0.336070 -0.945537 3.4669e-01
## time::-44 -0.514822    0.383762 -1.341516 1.8282e-01
## time::-43 -0.159374    0.352915 -0.451592 6.5255e-01
## time::-42 -0.026721    0.358904 -0.074452 9.4080e-01
## time::-41 -0.696081    0.284070 -2.450386 1.6025e-02 *
## time::-40 -0.327563    0.350936 -0.933397 3.5289e-01
## time::-39 -0.321154    0.244523 -1.313392 1.9209e-01
## time::-38 -0.225266    0.236530 -0.952377 3.4323e-01
## time::-37 -0.327997    0.251902 -1.302085 1.9591e-01
## time::-36 -0.466573    0.230839 -2.021206 4.5958e-02 *
## time::-35 -0.328069    0.244797 -1.340169 1.8326e-01
## time::-34 -0.807381    0.211300 -3.821022 2.3215e-04 ***
## time::-33 -0.364063    0.240569 -1.513345 1.3338e-01
## time::-32 -0.375912    0.242135 -1.552487 1.2374e-01
## time::-31 -0.302651    0.259664 -1.165550 2.4660e-01
## time::-30 -0.546525    0.235403 -2.321660 2.2301e-02 *
## time::-29 -0.611490    0.248401 -2.461703 1.5557e-02 *
## time::-28 -0.219903    0.238037 -0.923820 3.5783e-01
```

```

## time::-27 -0.418089 0.240952 -1.735155 8.5825e-02 .
## time::-26 -0.388816 0.239024 -1.626680 1.0698e-01
## time::-25 -0.616607 0.228697 -2.696170 8.2439e-03 **
## time::-24 -0.322887 0.224294 -1.439566 1.5314e-01
## time::-23 -0.402332 0.240051 -1.676027 9.6888e-02 .
## time::-22 -0.326878 0.219629 -1.488321 1.3984e-01
## time::-21 -0.375812 0.218353 -1.721125 8.8352e-02 .
## time::-20 -0.628534 0.222049 -2.830613 5.6269e-03 **
## time::-19 -0.252472 0.227075 -1.111841 2.6890e-01
## time::-18 -0.537204 0.235685 -2.279333 2.4794e-02 *
## time::-17 -0.482935 0.227290 -2.124748 3.6098e-02 *
## time::-16 -0.526256 0.259545 -2.027609 4.5287e-02 *
## time::-15 -0.286743 0.234811 -1.221166 2.2492e-01
## time::-14 -0.565931 0.230128 -2.459199 1.5660e-02 *
## time::-13 -0.531043 0.252331 -2.104553 3.7863e-02 *
## time::-12 -0.174406 0.218246 -0.799122 4.2613e-01
## time::-11 -0.798886 0.237953 -3.357327 1.1178e-03 **
## time::-10 -0.293696 0.245444 -1.196593 2.3432e-01
## time::-9 -0.478051 0.262249 -1.822891 7.1338e-02 .
## time::-8 -0.657397 0.236388 -2.781006 6.4881e-03 **
## time::-7 -0.244442 0.233263 -1.047923 2.9723e-01
## time::-6 -0.612850 0.235379 -2.603669 1.0642e-02 *
## time::-5 -0.701941 0.240264 -2.921536 4.3148e-03 **
## time::-4 -0.234886 0.241750 -0.971608 3.3361e-01
## time::-3 -0.378554 0.237030 -1.597074 1.1344e-01
## time::-2 -0.258616 0.235704 -1.097208 2.7521e-01
## time::0 1.067801 0.271647 3.930833 1.5698e-04 ***
## time::1 1.391231 0.254041 5.476403 3.2950e-07 ***
## time::2 0.982791 0.228663 4.297983 4.0381e-05 ***
## time::3 1.524969 0.244207 6.244576 1.0663e-08 ***
## time::4 1.533971 0.231510 6.625931 1.8175e-09 ***
## time::5 1.219927 0.251848 4.843896 4.7086e-06 ***
## time::6 1.489349 0.235819 6.315646 7.6909e-09 ***
## time::7 1.618923 0.262719 6.162173 1.5547e-08 ***
## time::8 0.924054 0.243691 3.791912 2.5721e-04 ***
## time::9 1.146870 0.235680 4.866218 4.2994e-06 ***
## time::10 1.374709 0.246886 5.568190 2.2103e-07 ***
## time::11 1.545583 0.244717 6.315797 7.6855e-09 ***
## time::12 1.763093 0.247230 7.131399 1.6506e-10 ***
## time::13 1.427923 0.228802 6.240854 1.0847e-08 ***
## time::14 1.578398 0.245624 6.426065 4.6163e-09 ***
## time::15 1.383354 0.205822 6.721112 1.1620e-09 ***
## time::16 1.386499 0.259497 5.343032 5.8525e-07 ***
## time::17 1.734093 0.244447 7.093938 1.9754e-10 ***
## time::18 1.919753 0.255990 7.499319 2.7895e-11 ***
## time::19 2.107906 0.220298 9.568412 9.5643e-16 ***
## time::20 1.693393 0.247098 6.853125 6.2259e-10 ***
## time::21 1.855829 0.251555 7.377428 5.0402e-11 ***
## time::22 2.242139 0.245811 9.121394 9.0174e-15 ***
## time::23 2.034422 0.244244 8.329460 4.7010e-13 ***
## time::24 2.069845 0.243376 8.504727 1.9666e-13 ***
## time::25 2.146262 0.257420 8.337589 4.5151e-13 ***
## time::26 2.228641 0.263677 8.452146 2.5549e-13 ***
## time::27 2.163605 0.254458 8.502804 1.9855e-13 ***

```

```

## time::28 2.304640 0.214881 10.725176 < 2.2e-16 ***
## time::29 2.206186 0.247421 8.916711 2.5140e-14 ***
## time::30 2.340212 0.262056 8.930198 2.3499e-14 ***
## time::31 2.340972 0.224114 10.445439 < 2.2e-16 ***
## time::32 2.767920 0.265952 10.407608 < 2.2e-16 ***
## time::33 2.400929 0.238276 10.076230 < 2.2e-16 ***
## time::34 2.414686 0.260145 9.282067 4.0278e-15 ***
## time::35 2.453759 0.262276 9.355631 2.7843e-15 ***
## time::36 2.135726 0.245806 8.688675 7.8598e-14 ***
## time::37 2.588958 0.231675 11.174936 < 2.2e-16 ***
## time::38 2.461614 0.236738 10.398043 < 2.2e-16 ***
## time::39 2.530024 0.240318 10.527801 < 2.2e-16 ***
## time::40 2.311900 0.251202 9.203347 5.9786e-15 ***
## time::41 3.834899 0.343612 11.160550 < 2.2e-16 ***
## time::42 3.977278 0.348645 11.407811 < 2.2e-16 ***
## time::43 3.316922 0.389966 8.505673 1.9574e-13 ***
## time::44 3.916513 0.369720 10.593183 < 2.2e-16 ***
## time::45 3.896915 0.378241 10.302744 < 2.2e-16 ***
## time::46 3.856503 0.422527 9.127227 8.7575e-15 ***
## time::47 3.727200 0.365424 10.199650 < 2.2e-16 ***
## time::48 4.269091 0.376386 11.342314 < 2.2e-16 ***
## time::49 3.886414 0.367210 10.583620 < 2.2e-16 ***
## time::50 3.981524 0.352868 11.283331 < 2.2e-16 ***
## time::51 4.388748 0.350345 12.526932 < 2.2e-16 ***
## time::52 4.404384 0.395315 11.141449 < 2.2e-16 ***
## time::53 4.228022 0.362436 11.665560 < 2.2e-16 ***
## time::54 4.174595 0.381403 10.945353 < 2.2e-16 ***
## time::55 4.517951 0.324688 13.914768 < 2.2e-16 ***
## time::56 4.245928 0.406686 10.440303 < 2.2e-16 ***
## time::57 4.353497 0.331246 13.142781 < 2.2e-16 ***
## time::58 4.699505 0.335418 14.010874 < 2.2e-16 ***
## time::59 4.507409 0.306667 14.698060 < 2.2e-16 ***
## time::60 4.321073 0.359144 12.031582 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.988262 Adj. R2: 0.815953
## Within R2: 0.398861

```